Chapter 5: SME Knowledge Gaps: What Regional SMEs Still Do Not Know About AI

1. The narrow question

- Many regional SMEs in the Netherlands are adopting sustainable commuting options like cycling or public transport to reduce car usage. However, they face challenges such as practicality issues and operational inefficiencies that hinder full adoption of these solutions.
- What specific knowledge gaps do regional SME owners need to address before committing budget for AI tools aimed at improving their operations?

2. The translation

- Regional SMEs in the Netherlands recognize the benefits of reducing car usage through sustainable commuting options like cycling or public transport, but they also acknowledge practical challenges and operational inefficiencies that prevent full adoption.
- Closing these knowledge gaps is crucial to ensure that AI solutions are deployed effectively, avoiding costly missteps and ensuring realistic ROI.

3. Illustrative use case (CRISP-DM)

Business understanding

- A regional SME in the Netherlands operates a seasonal business with limited internal IT staff. They need an automated logistics platform to streamline their operations during peak seasons but struggle with integrating it into their existing ERP system, leading to delays and inefficiencies.

Data understanding

- Historical data from the ERP shows inconsistent entries (e.g., missing fields for delivery addresses or item details), which could cause errors in the AI-generated recommendations.
- Inconsistent historical data also includes outdated records that do not reflect current business practices, such as obsolete pricing information or incorrect inventory levels.

Data preparation

- To prepare the data for use with an AI tool, SME owners need to spend 18-24 months cleaning and standardizing their ERP records. This involves identifying missing fields, correcting inconsistencies, and ensuring all relevant data is present.
- The process includes manual spot-checks of historical records, which can be time-consuming but necessary to ensure the accuracy of the AI model.

Modelling

- Generic off-the-shelf models often fail when deployed on uncleaned data. For example, if a delivery address field is missing from an order record, the AI tool might generate recommendations based on incorrect or incomplete information.
- This can lead to errors in inventory management, customer service issues, and ultimately financial losses.

Evaluation

- To mitigate risks associated with deploying AI tools prematurely, SMEs should set strict confidence thresholds for their models. For instance, if a recommendation has less than 85% confidence, it must be reviewed manually by human experts.
- This ensures that critical decisions are not made based on unreliable data and helps maintain the integrity of the system.

Deployment

- Deploying AI tools without addressing underlying data issues can lead to guesswork and potential failures. For example, if a model is deployed with incomplete or inaccurate historical data, it may generate recommendations that do not align with current business realities.
- This could result in operational disruptions, financial losses, and damage to the company's reputation.

4. Recommended tooling

- Low-cost tools for SMEs include staff AI usage surveys to identify existing AI solutions and their effectiveness.
- Free readiness assessment checklists can help SMEs evaluate their data quality and infrastructure needs before

investing in more advanced tools.

- Manual spot-checks of historical ERP records are a practical way to ensure the accuracy of the data used by AI models.

5. Responsible AI and Trust

- The Data Boundary Rule specifies that no raw ERP exports, client identifiers, or proprietary pricing should be pasted into external public Large Language Models (LLMs).
- This rule helps protect sensitive information while ensuring compliance with relevant regulations such as GDPR and the EU's AI Act.

6. First five steps for an owner

21. Informally list which AI tools staff already use.
22. Select one specific operational bottleneck, defining what must be true to trust an AI tool with it.
23. Conduct a spot check on historical records in the ERP for missing or invalid attributes.
24. Consult an independent trusted advisor on real ROI achieved by comparable firms.
25. Formulate a documented gap list as the direct input for choosing a use case in Chapter 6.

7. Sources

- The findings from field data and interviews are used to understand regional SMEs' specific challenges with AI adoption, while secondary reports (CBS, Eurostat, OECD) provide broader context on industry trends and regulatory requirements.

Chapter 6: Practical AI Use Cases for Regional SMEs

1. Executive Summary & Context

- [Company_1] B.V., a Dutch regional SME operating in the logistics sector, saw significant revenue growth of 14.8% in Q3 2026 compared to the same period last year.
- The company's growth was driven by launching an automated logistics platform and expanding into the Scandinavian market. Key performance indicators (KPIs) showed a customer retention rate increase from 90.2% to 92.3%, with operational costs rising slightly due to investments in cloud infrastructure and hiring AI specialists.

2. Framework: The 3 AI Maturity Tiers for SMEs

Tier 1: Off-the-Shelf Generative & Visual Tools (Rapid Deployment)

- Description: Plug-and-play SaaS solutions like automated email routing, lightweight marketing tools, or visual asset generators.
- Primary Goal: Eliminate administrative friction and lower time-to-market.

Tier 2: Workflow & Process Automation (Task-Level Optimization)

- Description: Embedded tools integrated with ERP/PIM systems for tasks such as data cleaning, translation, and localized workflows.
- Primary Goal: Save labor hours, eliminate manual spreadsheet entries, and reduce material waste.

Tier 3: Predictive Operations & Demand Balancing (Integrated Intelligence)

- Description: Fine-tuned predictive ML models for demand forecasting, automated reorders, and stock balancing with human-in-the-loop oversight.
- Primary Goal: Prevent stockouts and balance seasonal spikes with human intervention.

3. Comparative Benchmarks & Operational Performance

Maturity Tier	Implementation Focus	Operational Impact	Key SME ROI Anchor / KPI
Tier 1	Off-the-Shelf Tools	Administrative friction reduced, time-to-market lowered	Customer retention rate increase, operational efficiency gains
Tier 2	Workflow Automation	Labor hours saved, manual entry eliminated, material waste reduction	Operational cost savings, productivity improvements
Tier 3	Predictive Analytics	Stock balancing and demand forecasting accuracy improved, stockouts prevented	Net profit margin improvement, supply chain efficiency

4. Regional Perspectives: Three AI Economies

United States:

- Capital-intensive models focused on rapid market dominance and proprietary platforms.
- High investment in technology to achieve quick market penetration.

China:

- State-directed industrial networks emphasizing speed, manufacturing throughput, and shop-floor automation.
- Vertical integration for efficient production processes.

Europe:

- SME Sovereignty & Sustainability Route prioritizing human-in-the-loop governance, worker autonomy, and strict compliance with regulations like GDPR and the EU AI Act.
- Emphasis on sustainable practices and ethical AI deployment.

5. Quick-Start Action Sheet: SME AI Implementation Roadmap

Phase 1: Assessment & Data Audit (Weeks 1-2)

- Conduct an audit of ERP records, enforce local anonymization rules, and define key performance indicators (KPIs).

Phase 2: Tier 1 & Tier 2 Pilot Deployment (Weeks 3-4)

- Deploy plug-and-play tools for administrative tasks, automate workflows with embedded systems, and verify data integrity through spot-checks.

Phase 3: Workflow Integration & Human Safeguards (Weeks 5-6)

- Establish failure hooks to handle errors automatically, train internal IT staff on AI use, and ensure compliance with regional regulations.